Multimodal AI-Based Healthcare Diagnostic & Clinical Decision Support System (CDSS)

Project Blueprint — Dataset, Architecture & Roadmap

1. Project Overview

Goal: Build a system that takes in more than one type of patient data — for example, a medical image (X-ray/MRI/CT), a radiology report or clinical notes (text), and structured clinical variables (age, vitals, lab values) — and fuses them to produce a diagnostic prediction plus a human-readable explanation, positioned as a decision-support tool for clinicians rather than a fully autonomous diagnoser.

Why "multimodal" matters here: Real clinical decisions are rarely made from a single data type. A chest X-ray alone is ambiguous without the patient's history and the radiologist's notes. A model that fuses image + text (and optionally tabular data) more closely mirrors real diagnostic reasoning and is a strong differentiator for a portfolio/interview project.

2. Recommended Kaggle Dataset(s)

Primary recommendation — Indiana University Chest X-Rays (image + report pairs)

Link: https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university

- What it is: ~7,470 chest X-ray images (frontal + lateral) paired with ~3,955 de-identified radiology reports (Indication, Comparison, Findings, Impression sections).
- Why it fits: genuinely multimodal, paired image–text data (not two unrelated datasets glued together); the standard "IU X-ray" academic benchmark, so there's a large body of published work to compare against.
- Primary use: image encoder (CNN/ViT) + text encoder (clinical BERT) fused for (a) disease/finding classification, (b) automatic impression generation, or (c) both.

Complementary datasets (to round out the multimodal + CDSS story)

Dataset	Kaggle Link	Modality it adds	Suggested Role
Heart Disease Dataset (Cleveland/UCI mirror)	kaggle.com/datasets/redwankarimsony/heart-disease-data	Structured/tabular clinical variables	Second pipeline for structured-data risk scoring; fused as auxiliary features
COVID-19 Radiography Database	kaggle.com/datasets/tawsifurrahman/covid19-radiography-database	Chest X-ray images (COVID/Normal/Pneumonia/Opacity)	Extra imaging data to pretrain/stress-test the image encoder
Chest X-Ray Images (Pneumonia)	kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia	Chest X-ray images	Additional labeled data / transfer learning source
MIMIC-CXR (PhysioNet, credentialed)	physionet.org/content/mimic-cxr/	Large-scale image+report	Optional stretch dataset for production-scale results

Note: true patient-level multimodal datasets (image + text + structured EHR, same patient) are rare on open platforms due to privacy. The standard academic workaround — used in the IU X-ray/MIMIC-CXR literature — is to pair each image with its own report text, and treat structured data as a separate track or a simulated/synthetic add-on. That approach is recommended here.

3. System Architecture

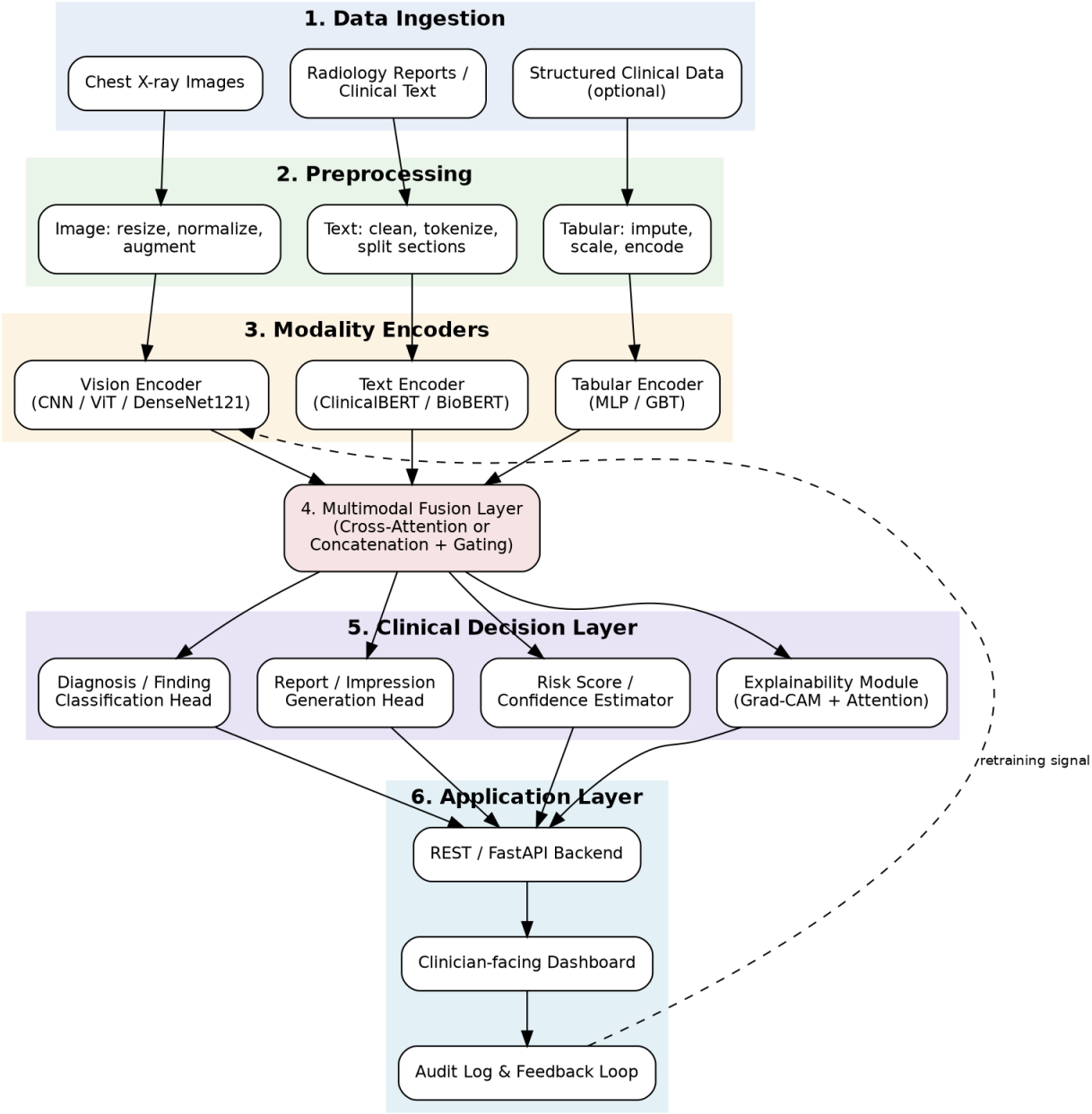


Figure 1: End-to-end multimodal CDSS architecture

Architecture notes

- Vision encoder: start with a pretrained DenseNet121 or ResNet50 (widely used baseline for chest X-rays, e.g. CheXNet), or a ViT for a more modern story.
- Text encoder: ClinicalBERT / Bio_ClinicalBERT (pretrained on MIMIC notes) works well out-of-the-box for radiology-style text.
- Fusion: start simple — concatenate image and text embeddings, then an MLP. Once that works end-to-end, upgrade to cross-attention (image tokens attend to text tokens and vice versa) as a stretch goal.
- Explainability is not optional for a CDSS — clinicians won't trust a black box. Grad-CAM on the image branch and attention-weight visualization on the text branch should be built in from the start.

4. Data Flow / Pipeline

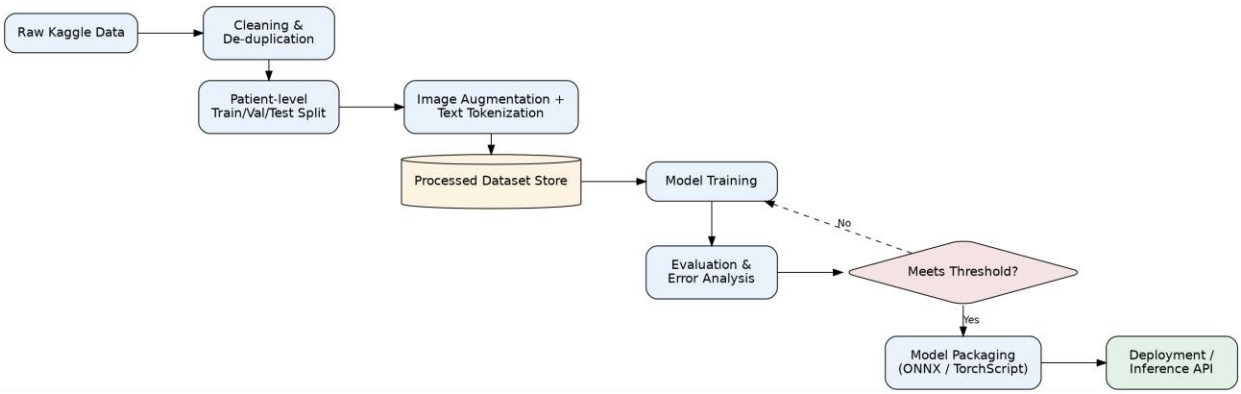


Figure 2: Data pipeline from raw Kaggle data to deployed model

Critical detail: split by patient ID, not by image — the IU X-ray dataset has multiple images per patient. Splitting randomly by image leaks patient information between train and test and inflates reported accuracy without meaning anything.

5. Model Training Roadmap (Guiding Steps)

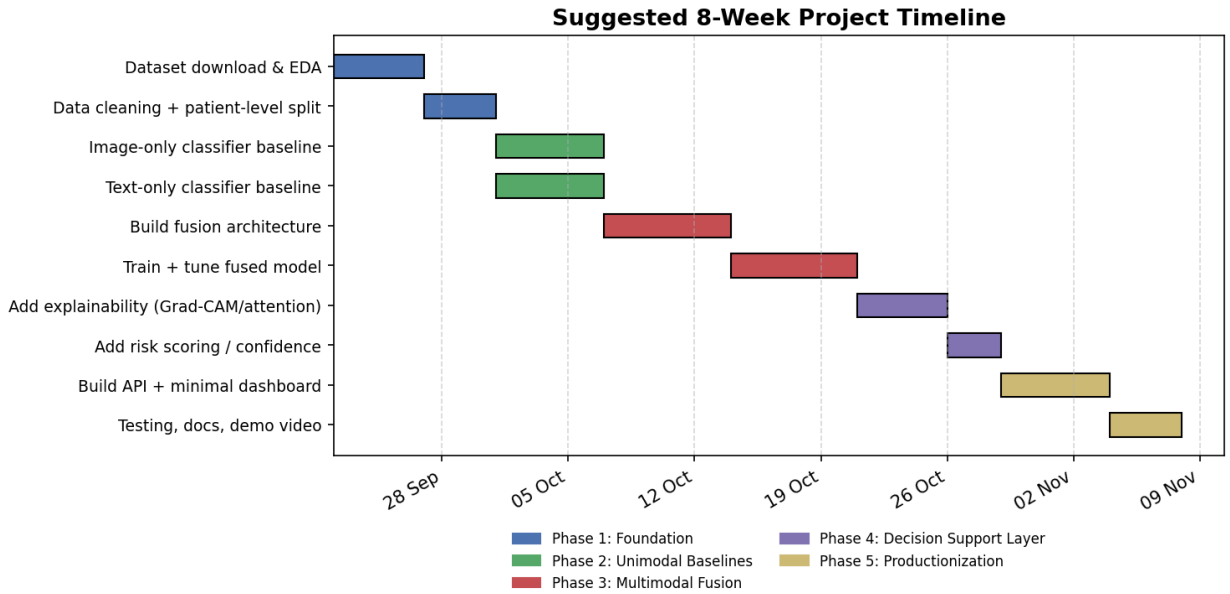


Figure 3: Suggested 8-week project timeline

Step-by-step guide

1. Problem framing. Decide the exact clinical task before writing code: e.g. "multi-label classification of 14 chest findings" (used in most IU X-ray papers) is the safest, most defensible scope. "Automatic report generation" is more impressive but harder to evaluate well — treat it as a stretch goal, not the MVP.

2. Exploratory Data Analysis (EDA). Check class balance (findings labels are usually heavily imbalanced), report length distribution, missing/duplicate images, and how many patients have more than one image.

3. Labeling strategy. IU X-ray reports are free text, not clean labels. Use the CheXpert labeler (open-source rule-based tool) to extract structured labels (e.g., Pneumonia: positive/negative/uncertain) from the Impression/Findings sections.

4. Patient-level train/val/test split (70/15/15 is standard in the literature) — never split at image level.

5. Unimodal baselines first. Train an image-only model and a text-only model separately to prove the multimodal model actually adds value.

6. Multimodal fusion. Start with simple concatenation fusion, then iterate to attention-based fusion if time allows.

7. Explainability layer. Grad-CAM for images, attention-weight heatmaps for text tokens — this turns a classifier into a decision support system.

8. Confidence/risk scoring. Output a calibrated probability, not just a label. Consider temperature scaling or Monte Carlo dropout for uncertainty estimates.

9. Evaluation. Use AUROC, F1, and precision/recall per finding (accuracy alone is misleading on imbalanced medical data). For report generation, add BLEU/ROUGE/METEOR plus qualitative review.

10. Serving layer. Wrap the trained model in a FastAPI backend; build a lightweight dashboard (Streamlit/React) where a clinician uploads an image + report and sees prediction + heatmap + confidence.

11. Ethical/limitations section. Explicitly document that this is a decision-support prototype, not a diagnostic device, dataset bias (single hospital network, demographic skew), and that it has not undergone clinical validation.

6. Suggested Tech Stack

Layer	Tools
Data handling	Python, Pandas, Pillow/OpenCV, NLTK/spaCy
Labeling	CheXpert labeler (rule-based NLP on reports)
Modeling	PyTorch (recommended) or TensorFlow/Keras
Vision backbone	torchvision DenseNet121/ResNet50, or timm for ViT
Text backbone	HuggingFace Transformers — Bio_ClinicalBERT
Experiment tracking	Weights & Biases or MLflow
Explainability	pytorch-grad-cam, attention visualization utilities
Serving	FastAPI + Uvicorn
Dashboard	Streamlit (fastest) or React (more polished)
Deployment	Docker container; Hugging Face Spaces or a cloud VM for a live demo

7. Repository Structure (suggested)

```
multimodal-healthcare-cdss/
├── data/
│   ├── raw/
│   ├── processed/
│   └── splits/                # patient-level train/val/test CSVs
├── notebooks/
│   ├── 01_eda.ipynb
│   ├── 02_labeling_chexpert.ipynb
│   └── 03_baseline_models.ipynb
├── src/
│   ├── data/                  # dataset loaders, preprocessing
│   ├── models/
│   │   ├── vision_encoder.py
│   │   ├── text_encoder.py
│   │   └── fusion_model.py
│   ├── explainability/
│   ├── training/
│   └── evaluation/
├── api/                        # FastAPI app
├── dashboard/                  # Streamlit/React app
├── reports/                    # figures, metrics, writeup
├── requirements.txt
└── README.md
```

8. Deployment / Serving View

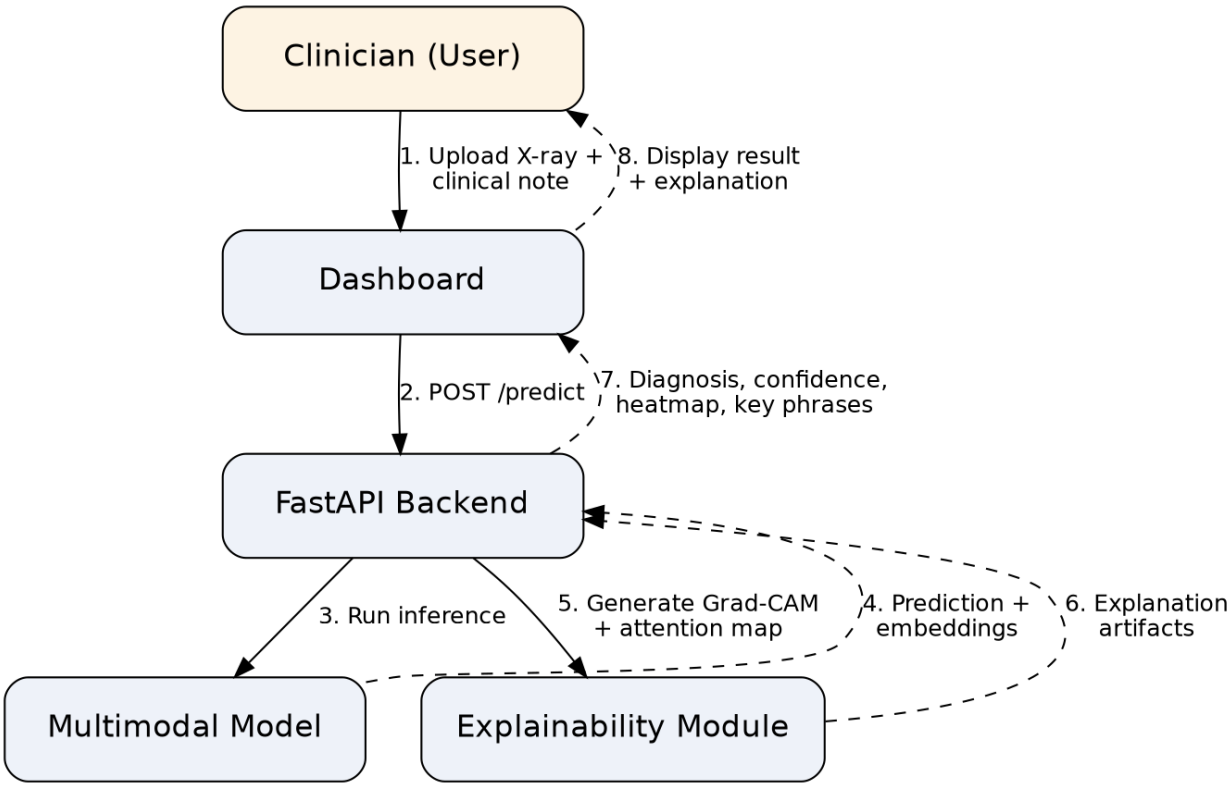


Figure 4: Inference request flow from clinician to model and back

9. Evaluation Checklist

- Patient-level split verified (no leakage)
- Class imbalance handled (weighted loss / focal loss / oversampling)
- Unimodal baselines reported alongside multimodal results
- AUROC, F1, precision, recall reported per finding — not just overall accuracy
- Confidence calibration checked (reliability diagram)
- Explainability outputs sanity-checked against a few known cases
- Limitations & ethical considerations section written
- Demo (dashboard + short video) prepared for presentation

10. Talking Points for Interviews / Presentations

- Why multimodal fusion beats unimodal (cite your own baseline comparison numbers).
- Why patient-level splitting matters (shows data leakage awareness — a common interview red flag if missed).
- How explainability was built in, not bolted on.
- Explicit acknowledgment that this is a decision-support prototype, not an autonomous diagnostic tool — shows clinical/ethical maturity.